Learning to Generate Text-guided 3D Canine-Object Interactions via Video Diffusion Priors

SIYU CHEN, Beijing Institute of Technology, China
ZAN WANG, Beijing Institute of Technology, China
HAO XU, Beijing Institute of Technology, China
MINGHE ZHANG, Beijing Institute of Technology, China
MINGTAO PEI, Beijing Institute of Technology, China
WEI LIANG✉, Beijing Institute of Technology, China

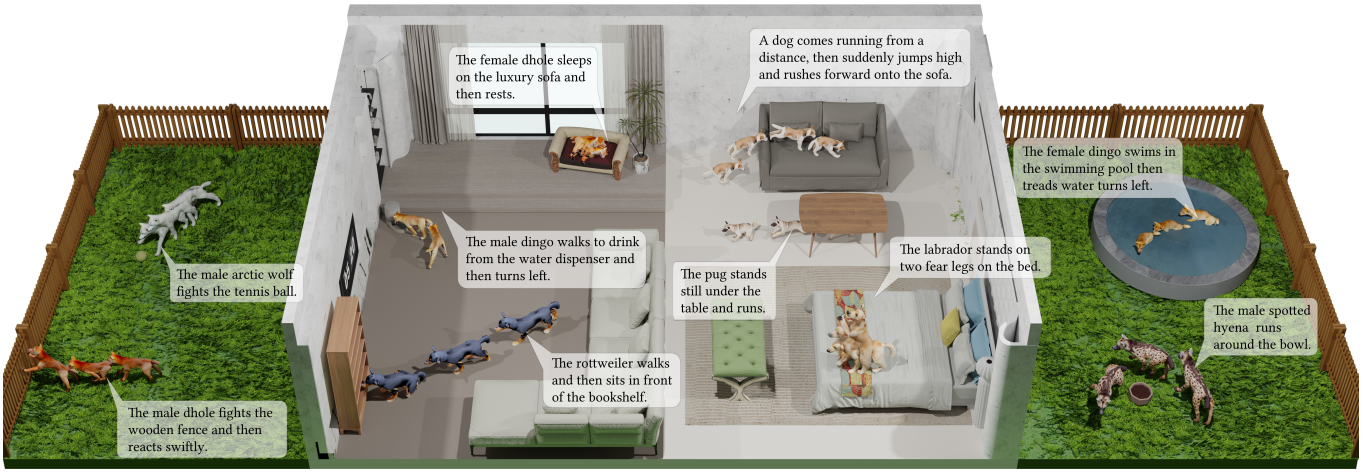


Fig. 1. We present **ZeroCOI**, a zero-shot learning framework for generating canine–object interactions from text prompts, trained without paired text–object–motion supervision. The right columns show representative interaction results, while the left columns demonstrate generalization to novel 3D object geometries and categories.

Generating realistic 3D Canine-Object Interaction from text is crucial for scalable asset production and immersive experiences in games and virtual reality. Despite substantial advances in Human-Object Interaction generation, extending this success to Canine-Object Interaction generation remains challenging, primarily due to the scarcity of text–object–motion pairings and heterogeneous skeletal topologies in existing animal motion datasets. To tackle these issues, we propose **ZeroCOI**, a zero-shot learning framework that synthesizes Canine-Object Interaction from text prompts without relying on paired text–object–motion supervision. Our key insight is to harness the rich semantic knowledge and implicit physics priors encoded in large-scale video diffusion foundation models trained on diverse in-the-wild Canine-Object Interaction patterns. **ZeroCOI** first learns canine motion priors from a topology-unified dataset using a text-to-motion diffusion model. It then decouples canonical motion generation from spatial interaction placement, introducing a control branch that injects semantic and spatial interaction cues via distribution-level video-domain Score Distillation Sampling while encouraging physical plausibility through geometric constraints. Extensive experiments demonstrate that **ZeroCOI** enables semantically aligned and physically plausible Canine-Object Interaction synthesis from text prompts, with robust generalization to unseen object categories and geometries. Furthermore, extended applications illustrate the scalability of **ZeroCOI** across various scenarios, including cross-species extensions, human-canine interactions, and scene-level interactions, demonstrating the tremendous potential for the broad community. *Code and dataset are available at https://github.com/BIT-PIE/ZeroCOI.*

CCS Concepts: • **Computing methodologies** → **Animation**; **Machine learning**.

Additional Key Words and Phrases: Canine Motion Generation, Canine-Object Interaction, Zero-Shot Learning

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Authors' Contact Information: Siyu Chen, 3120240925@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Zan Wang, wangzan@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Hao Xu, 3220251508@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Minghe Zhang, 3220251379@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Mingtao Pei, peimt@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Wei Liang, liangwei@bit.edu.cn, Beijing Institute of Technology, Beijing, China.



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1 Introduction

Realistic interactions between canines and 3D environmental objects are fundamental to computer graphics, virtual and augmented reality (VR/AR), and embodied AI. Despite its importance, research on text-guided Canine-Object Interaction (COI) generation remains in its infancy, in stark contrast to the rapid progress achieved in Human-Object Interaction (HOI).

Rapid progress in HOI has been fueled by a well-established ecosystem of foundational resources, including large-scale public datasets capturing diverse HOI [Bhatnagar et al. 2022; Li et al. 2023; Taheri et al. 2020; Xu et al. 2025], unified parametric body models [Loper et al. 2015; Pavlakos et al. 2019], and standardized benchmarks [Kulkarni et al. 2024; Xu et al. 2023]. These foundations enable HOI research to move beyond isolated actions toward increasingly complex scenarios, *e.g.*, language-conditioned scene-aware interactions [Jiang et al. 2024b; Wang et al. 2024], long-horizon interaction planning [Wang et al. 2025c; Xiao et al. 2024], and multi-modal conditional generation [Chhatre et al. 2024; Zhang et al. 2024a].

However, such an ecosystem is largely absent in text-guided COI. The primary limitation is the lack of paired data that jointly aligns textual descriptions, environmental objects, and corresponding canine interactive motions. Moreover, existing animal motion datasets [Wang et al. 2025d; Yang et al. 2024; Zhang et al. 2024b] offer limited behavioral diversity, are fragmented across sources and species, and exhibit heterogeneous skeletal representations with inconsistent topologies. As a result, directly transferring end-to-end text-guided HOI generation methodologies [Jiang et al. 2024b; Wang et al. 2024] to COI domain remains intractable.

Given the lack of paired text–object–motion data, zero-shot learning has emerged as a promising paradigm for text-guided interaction synthesis. Recent work on zero-shot HOI synthesis leverages external generative priors from pre-trained image/video diffusion models [Cao et al. 2025; Kim et al. 2025; Li et al. 2026] and large language models (LLMs) for high-level semantics and contact reasoning [Gui et al. 2026; Guo et al. 2025; Xu et al. 2024]. In contrast to methods that rely on instance-specific optimization or inference-time adaptation, we leverage external generative priors as training-time interaction supervision, enabling interaction knowledge to be learned once during training and effectively generalized to unseen inputs.

Specifically, we incorporate feedback from large-scale video diffusion models [Chen et al. 2024; Wang et al. 2023b] into a unified, object-aware generative model during training via distribution-level video-domain Score Distillation Sampling (SDS) [Poole et al. 2023; Singer et al. 2023]. Benefiting from extensive paired text–video data, the feedback effectively assesses whether synthesized interactions are semantically consistent with the input text (*i.e.*, *semantic knowledge*) and physically plausible with respect to the interacting objects (*i.e.*, *physics priors*). The semantic priors capture both spatial relationships (*e.g.*, "near" versus "on" an object) and interaction-specific motion patterns (*e.g.*, "jump onto the bed" versus "jump on the ground"), while the physics priors penalize physically implausible interactions such as object penetration.

Building on this insight, we introduce **ZeroCOI**, a zero-shot learning framework for synthesizing Canine-Object Interaction (COI) from text prompts without relying on paired text–object–motion supervision. We first pre-train a text-conditioned diffusion model [Tevet et al. 2023] on the canonical text-to-canine-motion dataset, enabling it to learn a canine motion prior aligned with textual descriptions. To incorporate object awareness, we further finetune the model with an augmented object-aware control branch that injects semantic interaction constraints via video-domain SDS, complemented by geometric constraints to encourage physical plausibility. Furthermore, we unify multiple datasets [Shooter et al. 2024; Wang et al. 2025d; Yang et al. 2024] into a consistent SMAL skeletal topology [Rueegg et al. 2022; Zuffi et al. 2017], resulting in DogML3D, a large-scale text-to-canine-motion dataset comprising 6,724 motion sequences and 13,392 textual descriptions across 39 canine species with diverse text-aligned motion patterns.

We evaluate **ZeroCOI** on DogML3D using a curated set of interaction object models. Extensive experiments demonstrate that our method generates diverse, physically plausible, and semantically consistent COI motions under language conditions, while generalizing effectively to unseen object categories and geometries. Beyond the canine-object interaction setting, we further explore various application scenarios, including cross-species extensions, human-canine interactions, and scene-level interactions.

Our contributions are threefold:

- We introduce zero-shot text-guided COI generation, where paired text–object–motion data is unavailable. To the best of our knowledge, we are the first to explicitly study this problem.
- We propose **ZeroCOI**, a unified zero-shot framework leveraging video diffusion priors as training-time supervision, and construct DogML3D by unifying existing canine motion datasets into a consistent SMAL representation.
- Through extensive experiments, we demonstrate that **ZeroCOI** generates diverse COI from text and generalizes robustly to unseen object categories and geometries.

2 Related Works

Animal motion generation remains challenging due to data scarcity and diverse skeletal structures, with prior work leveraging internet videos via reconstruction or weak supervision [Cheng et al. 2023; Kapon et al. 2024; Sun et al. 2024; Yang et al. 2025], augmenting data through one-to-many motion synthesis [Raab et al. 2024; Wang et al. 2025a; Zhang et al. 2024b], and improving generalization via topology-agnostic representations [Chen et al. 2025; Gat et al. 2025; Lee et al. 2025]. Another line of work explores cross-category motion transfer [Li et al. 2024; Zhang et al. 2025; Zhao et al. 2024], which aligns existing source trajectories across heterogeneous morphologies rather than generating motions from scratch. Despite these advances, existing methods primarily focus on isolated motion and do not explicitly model object-centric interactions. In parallel, human-object interaction (HOI) generation benefits from large paired datasets [Jiang et al. 2024a,b; Liu et al. 2025; Xu et al. 2025], enabling text-guided interaction modeling [Song et al. 2024; Wang et al. 2025b], while recent zero-shot approaches reduce data dependency by leveraging diffusion priors or LLM-guided semantics [Cao et al. 2025; Gui et al. 2026; Guo et al. 2025; Kim et al. 2025; Xu et al. 2024; Zhu et al. 2025], though they often rely on

instance-specific optimization and lack scalability. Separately, visual diffusion models and their score-based supervision have been widely used for 3D/4D generation [Bahmani et al. 2024; Ren et al. 2023; Singer et al. 2023; Sun et al. 2025] via SDS [Poole et al. 2023]. ProlificDreamer [Wang et al. 2023a] further generalizes SDS to Variational Score Distillation(VSD), modeling 3D scene parameters as a distribution instead of a single deterministic representation, while still relying on per-scene optimization rather than learning reusable priors. Similarly, distribution matching distillation [Yin et al. 2024] aligns generator distributions with diffusion models but is mainly designed for efficient image synthesis.

3 Method

3.1 Overview

Problem Formulation. Given a language description $\mathcal{L} = \{w_m\}_{m=1}^{M}$ comprising M tokenized words and a target object $O = (V_o, F_o)$ represented as a triangular mesh with vertices $V_o \in \mathbb{R}^{N_o \times 3}$ and faces $F_o \in \mathbb{N}^{N_f \times 3}$, our goal is to synthesize a canine motion sequence $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{F}$ that is semantically consistent and physically plausible with the given conditions. F is the number of motion frames.

We use D-SMAL [Rueegg et al. 2022] to represent the canine body in each frame, *i.e.*, $\mathbf{x}_i$, which is parameterized by shape coefficients $\beta \in \mathbb{R}^{30}$, joint rotations $\theta \in \mathbb{R}^{6N}$ in 6D representation, global orientation $r \in \mathbb{R}^6$, and global translation $t \in \mathbb{R}^3$. To handle the wide variation in canine body sizes, we further introduce a scaling factor s. Given these parameters, the posed 3D mesh is formulated as $\mathcal{V}, \mathcal{F} = \mathcal{M}(\beta, \theta, r) \cdot s + t$, where $\mathcal{V} \in \mathbb{R}^{3889 \times 3}$ are the mesh vertices, $\mathcal{F} \in \mathbb{N}^{7774 \times 3}$ denotes the triangular faces, and $\mathcal{M}(\cdot)$ is the D-SMAL skinning function. Following HumanML3D [Guo et al. 2022], we adopt a redundant motion representation to facilitate diffusion-based modeling. Each pose in the motion is $\mathbf{x}_i = \left[\dot{r}^a, \dot{r}^x, \dot{r}^z, r^y, j^p, t, r, \theta, j^v, c^f\right]_i \in \mathbb{R}^{428}$, where $\dot{r}^a \in \mathbb{R}$ is root angular velocity along Y-axis; $(\dot{r}^x, \dot{r}^z \in \mathbb{R})$ are root linear velocities on XZ-plane; $r^y \in \mathbb{R}$ is root height; $j^p \in \mathbb{R}^{3N}$ and $j^v \in \mathbb{R}^{3N}$ are the local joints positions and velocities in root space; $c^f \in \mathbb{R}^4$ is binary foot ground contact labels.

Pipeline Overview. We propose **ZeroCOI**, a zero-shot learning framework for synthesizing COI from text prompts without requiring text–object–motion pairings, as illustrated in Fig. 2.

Our framework follows a two-stage learning procedure. In the *Text-to-motion Pretraining stage*, we pretrain a canonical text-to-motion diffusion model to capture a realistic distribution of canine motion independent of any specific object, serving as a strong and stable motion prior. In the *Object-aware Finetuning* stage, we finetune the pretrained motion generator to generate object-aware interaction. To inject interaction-specific priors without disrupting the learned motion distribution, we explicitly decouple canonical motion generation from world-space spatial placement and introduce an object-aware control branch to inject interaction-related knowledge. In the absence of text-object-motion pairings, we leverage pre-trained text-to-video diffusion models, *i.e.*, ZeroScope [cerspense 2023], as training-time semantic supervisors via distribution-level video-domain SDS, providing implicit preference signals that bias the model toward generating perceptually and physically plausible COI. We further augment motion-centric descriptions into interaction-centric instructions, providing the generative model with more precise interaction semantics, and render generated motions using precomputed textured canine meshes, ensuring that SDS gradients primarily reflect interaction semantics rather than appearance variations.

3.2 Motion Generation with Object-Aware Control

Text-to-Motion Pre-training. We learn the canine motion distribution using the diffusion model, which defines a forward Markov noising process $\{\mathbf{X}_t\}_{t=0}^{T}$. Specifically, starting from the clean motion $\mathbf{X}_0$, Gaussian noise is progressively injected, defined as $q(\mathbf{X}_t \mid \mathbf{X}_{t-1}) = \mathcal{N}\left(\sqrt{\alpha_t}\,\mathbf{X}_{t-1}, (1-\alpha_t)\mathbf{I}\right)$, where $\alpha_t \in (0,1)$ follows a predefined noise schedule. As t increases, the corrupted motion $\mathbf{X}_t$ approaches a standard Gaussian distribution. Motion generation is formulated as the reverse diffusion process that gradually denoises from $\mathbf{X}_T$ back to $\mathbf{X}_0$. T is the total diffusion steps. At each step t, the reverse transition is modeled as $p_\psi(\mathbf{X}_{t-1} \mid \mathbf{X}_t, \mathcal{L}, C) = \mathcal{N}\Big(\boldsymbol{\mu}_\psi(\mathbf{X}_t, t, \mathcal{L}, C), (1-\alpha_t)\mathbf{I}\Big)$, where $C = (\beta, s)$ denotes the morphology condition. Unlike human motion generation, which typically relies on a standardized body model, canine motion exhibits substantial breed- and morphology-dependent variations, making C essential for modeling diverse canine motion patterns.

Following [Tevet et al. 2023], we directly predict the clean motion $\hat{\mathbf{X}}_0 = G_\psi(\mathbf{X}_t, t, \mathcal{L}, C)$. The denoising model G_ψ is trained using the simple reconstruction objective, *i.e.*,

$$\mathcal{L}_{\text{MSE}} = \mathbb{E}_{\mathbf{X}_0, t}\left[\left\|\mathbf{X}_0 - \hat{\mathbf{X}}_0\right\|_2^2\right]. \tag{1}$$

Object-Aware Finetuning. To generate COI, we adopt a ControlNet-style control branch [Zhang et al. 2023] to finetune the frozen text-to-motion diffusion model G_ψ, enabling object-aware motion generation while preserving motion realism and text controllability. Specifically, the control branch mirrors the Transformer encoder of the denoising model backbone, takes object geometry as input, and connects back to the backbone at each Transformer layer through linear projection layers initialized to zero. Formally, with the object-aware control branch, the denoising process predicts the clean motion as $\hat{\mathbf{X}}_0 = G_{\psi,\psi^+}(\mathbf{X}_t, t, \mathcal{L}, C, O)$, where ψ denotes the frozen parameters of the pretrained text-to-motion diffusion backbone, and ψ^+ denotes the learnable parameters of the object-aware control branch.

We encode object geometry and COI features using the Basis Point Set (BPS) representation [Prokudin et al. 2019], a lightweight and compact encoding demonstrated by prior works [Li et al. 2023; Yi et al. 2024]. Given an object mesh $O = (V_o, F_o)$, we place it on the ground plane, align its centroid to the world XZ-origin, and orient it toward the negative Z-axis. We then sample 1024 points within a 1.5 m radius sphere centered at the object centroid to form the BPS $B \in \mathbb{R}^{1024 \times 3}$. We compute object geometry features by measuring the Euclidean distance from each sampled point to its nearest vertex on the object mesh, yielding $B_O \in \mathbb{R}^{1024}$. At denoising step t, given a noisy motion sequence $\mathbf{X}_t$ with F frames, we compute the minimum distance from each sampled point to any canine joint at each frame, producing frame-wise interaction features $B_D = [B^1, \cdots, B^F]$, where $B^i \in \mathbb{R}^{1024}$. The two features are concatenated and fed into an MLP to form a fused representation, which serves as input to the object-aware control branch.

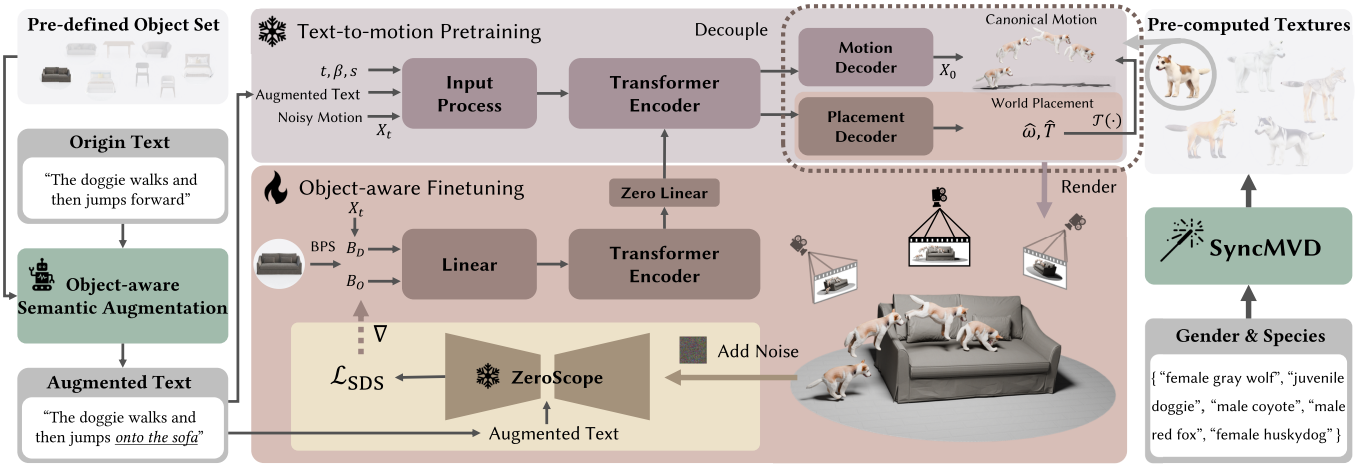


Fig. 2. **Overview of ZeroCOI**. We first augment motion-centric prompts with object-aware interaction descriptions to facilitate subsequent training. A text-to-motion diffusion model is pretrained to learn a robust canine motion prior, while object-aware finetuning leverages ZeroScope via video-domain SDS to supervise interaction semantics. To inject interaction-specific priors without disrupting the learned motion distribution, we explicitly decouple canonical motion generation from world-space spatial placement, and we render generated motions using precomputed mesh textures to ensure stable SDS supervision.

3.3 Semantic Injection via Video-domain SDS

Due to the absence of paired canine-object interaction data, we adopt video-domain SDS as a training-time semantic supervision, *i.e.*, leveraging ZeroScope as external priors.

In prior work [Bahmani et al. 2024; Singer et al. 2023], video-domain SDS is typically employed to iteratively optimize a 4D representation ϕ by computing gradients that align the rendered video $\mathbf{v}_0 = \mathcal{R}(\phi)$ given text prompt $\mathcal{L}$. The SDS gradient with respect to ϕ is given by

$$\nabla_\phi \mathcal{L}_{\text{SDS}} = \mathbb{E}_{t,\epsilon}\left[w(t)\ (\epsilon_\eta(\mathbf{v}_t; t, \mathcal{L}) - \epsilon)\ \frac{\partial \mathbf{v}_0}{\partial \phi} \right], \tag{2}$$

where $\mathbf{v}_t = \sqrt{\alpha_t}\mathbf{v}_0 + \sqrt{1-\alpha_t}\epsilon$ denotes the noisy video at diffusion timestep t, $\epsilon_\eta(\mathbf{v}_t; t, \mathcal{L})$ is the noise predicted by the pretrained text-to-video diffusion model conditioned on $\mathcal{L}$, and $w(t)$ is a timestep-dependent weighting factor.

This formulation treats SDS as an instance-level optimization method, refining a 4D representation ϕ independently for each prompt. In contrast, we extend video-domain SDS to the distribution level, optimizing a parametric motion generator rather than optimizing individual instances. Specifically, the predicted motion $\hat{\mathbf{X}}_0 = G_{\psi,\psi^+}(\mathbf{X}_t, t, \mathcal{L}, C, O)$ is converted into a D-SMAL mesh and jointly rendered with the target object O into a video $\hat{\mathbf{v}}_0 = \mathcal{R}(\hat{\mathbf{X}}_0, O)$ via differentiable rendering. The resulting distribution-level video-domain SDS gradient, which updates the object-aware control branch while keeping the pretrained backbone frozen, is formulated as

$$\nabla_{\psi^+} \mathcal{L}_{\text{SDS}} = \mathbb{E}_{t,\epsilon}\left[w(t)\ (\epsilon_\eta(\mathbf{v}_t; t, \mathcal{L}) - \epsilon)\ \frac{\partial \hat{\mathbf{v}}_0}{\partial \hat{\mathbf{X}}_0}\ \frac{\partial \hat{\mathbf{X}}_0}{\partial \psi^+} \right]. \tag{3}$$

Although $\mathcal{L}_{\text{SDS}}$ in Eq. 3 shares the same analytical form as instance-level SDS, its role here is fundamentally different. In our formulation, the pretrained text-to-video models induce a semantic gradient field that is projected into the motion space, biasing the distribution of canine motions rather than individual instances. Crucially, these updates are constrained to the motion-prior manifold defined by the frozen backbone, which means that injecting interaction semantics does not disrupt the learned motion distribution.

To enhance the stability and reliability of the distilled semantic gradients, we adopt a carefully designed differentiable rendering strategy. We leverage SyncMVD [Liu et al. 2024] to precompute textures for D-SMAL meshes from different canine breeds, and bind the corresponding mesh to the predicted pose at each frame during rendering. This strategy significantly stabilizes the rendered appearance across iterations and ensures that the SDS gradients primarily reflect semantic interaction cues rather than spurious texture variations.

3.4 Placement & Motion Decouple

A fundamental challenge arises from the mismatch between the canonical motion distribution learned from data and the diverse world-space configurations required for object-centric interactions. Specifically, the text-to-motion backbone is pretrained on motions represented in a canonical coordinate system, where all sequences are normalized to the origin and aligned to a fixed orientation. In contrast, interaction requires motions to exhibit semantically correct relative positions and orientations with respect to the target object. Directly enforcing world-space interaction constraints can introduce conflicting gradients that disrupt the learned motion prior, often degrading realism or inducing unnatural dynamics. This motivates an explicit decoupling of motion generation from world-space placement, allowing interaction objectives to be satisfied without compromising the underlying motion distribution.

We decouple the motion generation into canonical motion generation and global placement estimation. Formally, the denoising network with an object-aware control branch produces a shared latent representation $\mathbf{H}_t$, from which a motion head predicts a canonical clean motion sequence $\hat{\mathbf{X}}_0 = D_{\text{motion}}(\mathbf{H}_t)$, while a parallel

placement head predicts a global yaw rotation and 3D translation $(\hat{\omega}, \hat{\mathbf{T}}) = D_{\mathrm{RT}}(\mathbf{H}_t)$. The raw placement parameters are mapped to predefined ranges via a $\tanh(\cdot)$ activation, ensuring numerical stability and plausible interaction configurations. The final world-space motion is then obtained by applying a rigid transformation to the canonical motion,

$$\tilde{\mathbf{X}}_0 = \mathcal{T}(\hat{\mathbf{X}}_0; \hat{\omega}, \hat{\mathbf{T}}), \tag{4}$$

where $\mathcal{T}(\cdot)$ denotes a rotation around the vertical axis followed by 3D translation. This design encourages interaction constraints to be satisfied primarily through spatial alignment, while allowing the underlying motion structure to be modified only when sufficiently supported by strong semantic or physical evidence.

Consistent with this design, supervision is decoupled across motion and world spaces. In the canonical motion space, we apply the same reconstruction objective used in text-to-motion pretraining, computed against pseudo-targets (*i.e.*, canonical motions from DogML3D), to preserve motion styles. In contrast, interaction-driven objectives, including semantic injection via video-domain SDS and physical constraints, are applied to the transformed world-space motions.

3.5 Physical Constraints

To refine interaction geometry, we introduce a set of lightweight geometric and physical constraints that regularize the generated motions. These objectives complement the semantic and implicit physical supervision provided by video-domain SDS, improving contact precision and reducing penetration.

We encourage the canine to approach the target object surface at the last motion frame. Let $\mathbf{r}$ denote the root joint position at the last valid frame, and let $\{\mathbf{o}_i\}_{i=1}^{N}$ be sampled points on the object surface. The approach loss is defined as

$$\mathcal{L}_{\mathrm{app}} = \max\big(0,\ \mathrm{softmin}_i \|\mathbf{r} - \mathbf{o}_i\|_2 - d_{\mathrm{target}}\big), \tag{5}$$

where d_{target} specifies the desired proximity threshold. This loss serves as a weak spatial anchor that reduces the search space of object-relative placement, since multiple placements may satisfy video-domain SDS for spatially ambiguous prompts such as "near the bed". It also keeps the interaction pair near the rendered image center, which better matches the object-centric pretrained distribution of video diffusion models and improves early-stage training stability. Similar centering strategies are also adopted in prior SDS-based works [Li et al. 2025].

To prevent non-physical penetration between the canine body and the object, we leverage the object's signed distance field (SDF). Given canine vertices $\mathbf{v}_{t,j}$ at frame t and vertex index j, the penetration loss is formulated as

$$\mathcal{L}_{\mathrm{pen}} = \frac{1}{|\mathcal{P}|} \sum_{(t,j)\in\mathcal{P}} \max\big(0,\ -\mathrm{SDF}(\mathbf{v}_{t,j})\big), \tag{6}$$

where $\mathcal{P}$ denotes the set of penetrating vertices.

Finally, we impose adaptive contact constraints on foot joints. The contact loss is computed over the K smallest positive foot-to-surface distances across the motion sequence, preventing erroneous penalization of airborne phases such as jumping, *i.e.*,

$$\mathcal{L}_{\mathrm{con}} = \frac{1}{K} \sum_{k=1}^{K} \max\big(0,\ d_{(k)}\big), \tag{7}$$

where K is adaptively determined by the number of valid motion frames, and $d_{(k)}$ denotes the k-th smallest positive distance between a foot joint and the corresponding support surface.

4 Experiments

4.1 Evaluation Setup

4.1.1 Datasets. To support learning and evaluation of COI generation, we construct a motion-language dataset that addresses three data-level challenges: (i) heterogeneous skeletal representations that prevent direct aggregation of existing canine motion datasets; (ii) the absence of a unified motion representation in the parameter space of parametric canine models (e.g., SMAL), which hinders constructing a model-compatible motion prior; and (iii) the lack of object-aware interaction descriptions for learning conditioned motion behaviors. The full dataset is split into training, validation, and test sets with ratios of 0.80, 0.05, and 0.15, respectively.

To address skeletal heterogeneity, we consolidate multiple existing canine motion datasets [Shooter et al. 2024; Wang et al. 2025d; Yang et al. 2024] and retarget their diverse topologies to a shared kinematic structure. All motion data are mapped to a common SMAL representation [Rueegg et al. 2022; Zuffi et al. 2017] via an optimization-based retargeting procedure. We further apply mirroring for augmentation. This process yields **DogML3D**, a large-scale canine motion-language dataset with 6,724 motion sequences and 13,392 textual descriptions across 39 canine species.

To enable object-aware control, which is absent in existing canine motion datasets, we construct a predefined object set covering common interactions by combining 3D-FRONT assets [Fu et al. 2021] with procedurally generated objects from Tripo [AI 2024], yielding 29 object instances across 7 categories. We sample 1-3 objects randomly from the object set and prompt ChatGPT-4o [Achiam et al. 2023] to generate interaction-aware textual descriptions conditioned on both the original motion text and object category descriptions. This process yields 23,900 text descriptions and allows us to systematically introduce object-conditioned interaction semantics without requiring additional motion capture or manual annotation.

To evaluate generalization, we further construct a held-out object set consisting of 11 unseen objects, 5 objects sharing the same category but different geometries, and 6 objects from novel categories.

4.1.2 Baselines. Since zero-shot synthesis of canine-object interactions is unexplored, no existing methods are directly applicable. We therefore construct three adapted baselines representing complementary zero-shot paradigms: (i) video-based reconstruction-driven HOI synthesis; (ii) LLM-guided motion prior sampling; and (iii) text-driven 4D mesh animation, including

- **DAViD** [Kim et al. 2025] is a zero-shot HOI synthesis method that generates interaction videos using a pre-trained video diffusion model and lifts them to 3D via reconstruction. We adapt it to the canine motion domain by replacing its human motion prior with our pretrained canine motion prior before interaction-specific

Method	CLIP Score↑	CLIP Consistency↑	MD↑	MD_{local} ↑	$Pene_{\%obj}$ ↓	$Pene_{mean}$ ↓	$Pene_{max}$ ↓	FS↓
DAViD [Kim et al. 2025]	$23.5327^{\pm.020}$	$0.9697^{\pm.001}$	$\mathbf{0.5794}^{\pm.038}$	$0.1769^{\pm.017}$	$0.1024^{\pm.007}$	$0.0592^{\pm.011}$	$0.1185^{\pm.008}$	$0.4703^{\pm.012}$
TSTMotion [Guo et al. 2025]	$22.3618^{\pm.142}$	$0.9540^{\pm.001}$	$0.3964^{\pm.025}$	$0.0445^{\pm.002}$	$0.1247^{\pm.003}$	$0.1121^{\pm.024}$	$0.1534^{\pm.017}$	$0.5264^{\pm.012}$
AnimateAnyMesh [Wu et al. 2025]	$20.9025^{\pm.163}$	$\mathbf{0.9976}^{\pm.000}$	-	-	-	-	-	-
w/o video-domain SDS	$21.2033^{\pm.133}$	$0.9858^{\pm.000}$	$0.1713^{\pm.022}$	$0.1325^{\pm.017}$	$0.0776^{\pm.013}$	$0.0325^{\pm.004}$	$0.0726^{\pm.004}$	$0.0638^{\pm.016}$
w/o decoupling	$26.8792^{\pm.109}$	$0.9862^{\pm.001}$	$0.1790^{\pm.030}$	$0.0974^{\pm.009}$	$0.1027^{\pm.015}$	$0.1022^{\pm.003}$	$0.1426^{\pm.004}$	$0.2565^{\pm.015}$
w/o object-aware control branch	$25.3274^{\pm.164}$	$0.9878^{\pm.001}$	$0.2056^{\pm.028}$	$0.1242^{\pm.017}$	$0.0994^{\pm.016}$	$0.0345^{\pm.009}$	$0.0977^{\pm.003}$	$0.1150^{\pm.005}$
w/o approach	$28.1767^{\pm.237}$	$0.9872^{\pm.001}$	$0.5415^{\pm.025}$	$0.2528^{\pm.015}$	$0.0539^{\pm.017}$	$0.0221^{\pm.008}$	$0.0553^{\pm.002}$	$0.0482^{\pm.010}$
w/o physical	$28.1276^{\pm.165}$	$0.9877^{\pm.001}$	$0.5740^{\pm.021}$	$0.2466^{\pm.010}$	$0.0681^{\pm.021}$	$0.0284^{\pm.005}$	$0.0654^{\pm.006}$	$0.0642^{\pm.007}$
ZeroCOI (Ours)	$\mathbf{28.2139}^{\pm.169}$	$0.9886^{\pm.000}$	$0.3472^{\pm.036}$	$\mathbf{0.2546}^{\pm.012}$	$\mathbf{0.0535}^{\pm.013}$	$\mathbf{0.0218}^{\pm.001}$	$\mathbf{0.0523}^{\pm.002}$	$\mathbf{0.0439}^{\pm.009}$

Table 1. **Quantitative comparison on zero-shot canine-object interaction synthesis.** We report results on semantic alignment, motion quality and diversity, and physical plausibility metrics. All results are averaged over five runs, with 95% confidence intervals shown.

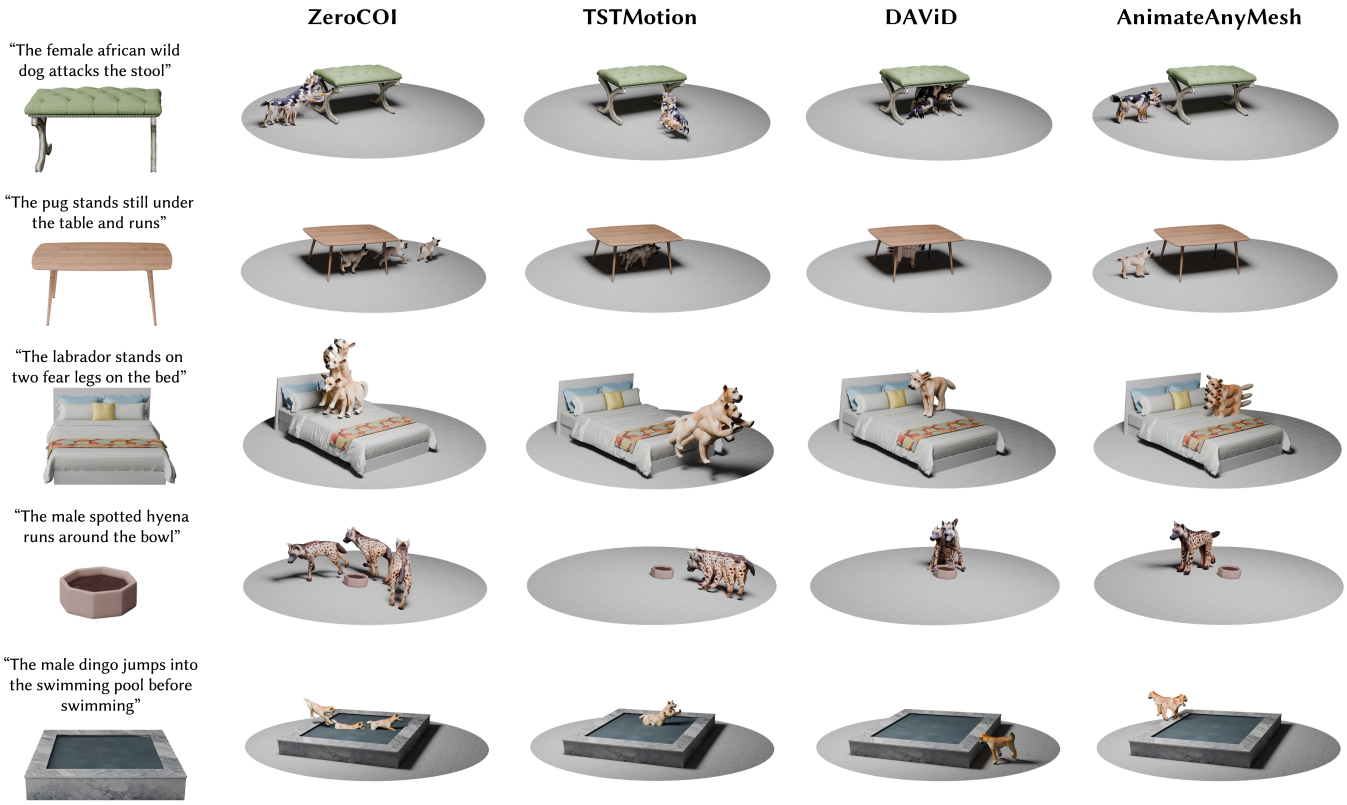


Fig. 3. **Qualitative comparison with baseline methods.** Given identical text prompts, baseline methods often produce conservative or semantically misaligned interactions, accompanied by artifacts such as object penetration and unstable foot contacts. **ZeroCOI** generates more expressive and semantically aligned interaction behaviors, maintaining stable contact and physically plausible motion trajectories.

finetuning, and its human motion reconstruction module [Shen et al. 2024] with *AnimalAvatar* [Sabathier et al. 2024], a state-of-the-art canine motion reconstruction method. Other components are kept unchanged.

- **TSTMotion** [Guo et al. 2025] formulates HOI generation as a guided motion synthesis process, where a LLM parses instructions and scenes to produce sparse joint-level waypoints that guide the sampling from a text-to-motion model. We adapt TSTMotion by redesigning the LLM prompts for canine interactions and retraining its motion prior [Xie et al. 2024] on DogML3D.
- **AnimateAnyMesh** [Wu et al. 2025] serves as a text-driven 4D mesh animation baseline. We initialize the canine mesh at a fixed distance (*i.e.*, $2m$) from the target object. As the method cannot enforce articulated kinematic consistency and introduce non-rigid deformations, we use it only for visual-level comparison.

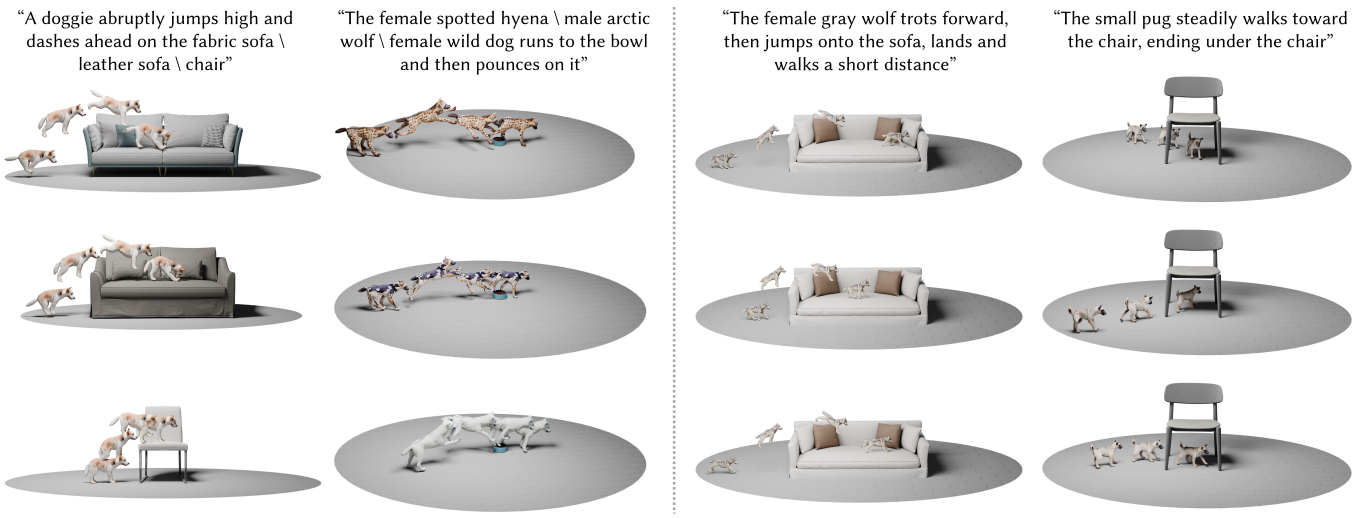


Fig. 4. **Adaptive and diverse interaction behaviors.** The left two columns show adaptation to varied object geometries and canine body shapes, while the right two columns show diverse motion dynamics and interaction trajectories under identical conditions.

Each baseline is modified minimally to operate in the canine domain, enabling fair comparison.

To analyze the contribution of individual components in **ZeroCOI**, we construct five ablative variants. **w/o video-domain SDS** removes the video-based semantic supervision during training, relying solely on text-to-motion learning and physical losses. **w/o decoupling** disables the decoupling between placement and motion generation, directly imposing interaction constraints in the motion space. **w/o object-aware control branch** removes the object-aware control branch and instead finetunes the pre-trained text-to-motion model by directly conditioning on object information using physical and video-domain SDS losses. **w/o approach** and **w/o physical** remove the approach loss and all physical constraints, respectively.

4.1.3 Metrics. Following the evaluation protocol of [Li et al. 2026], we adopt a set of complementary metrics: *CLIP Score*, measuring the semantic alignment between text prompts and rendered motion frames; *CLIP Consistency*, assessing temporal coherence via cosine similarity between adjacent frame embeddings; *Motion Diversity (MD)* and *Root-Local Motion Diversity (MD_{local})*, computed in the world and root-local coordinate systems as the average pairwise joint-wise Euclidean distance among multiple generated sequences with identical inputs; and *Physical Plausibility*, evaluated using penetration-related metrics including $Pene_{\%obj}$, $Pene_{mean}$, and $Pene_{\max}$ as well as the *Foot Sliding (FS)* score.

4.2 Quantitative Evaluation

We report quantitative comparisons on the zero-shot canine-object interaction synthesis task in Tab. 1. All evaluations are repeated five times to ensure robustness, with 95% confidence intervals. Overall, **ZeroCOI** demonstrates clear improvements over all baselines on both semantic alignment and physical plausibility metrics. Specifically, **ZeroCOI** achieves the highest *CLIP Score* while maintaining strong temporal consistency, indicating improved semantic alignment without degrading motion coherence. For motion diversity, while some baselines obtain higher world-space MD, their MD_{local} scores decrease substantially, suggesting that their diversity is largely influenced by unstable global placements. In contrast, **ZeroCOI** achieves the highest $\mathrm{MD}_{\mathrm{local}}$, indicating that it can produce diverse internal motion dynamics under identical inputs. Consistent gains are also observed in physical plausibility, where **ZeroCOI** achieves the lowest object penetration and foot sliding across all evaluated metrics.

We further analyze ablation variants of **ZeroCOI** in Tab. 1. Removing video-domain SDS supervision degrades both semantic alignment and physical plausibility, confirming its role in shaping a semantically and physically grounded interaction prior. Disabling the decoupling strategy increases penetration and reduces motion stability and diversity, while the object-aware control branch ablation consistently underperforms the full model; these results validate our two-stage, decoupled design for robust zero-shot interaction synthesis. Removing the approach loss increases world-space MD while leaving other metrics nearly unchanged, indicating its effect in stabilizing global placement. Removing all physical constraints slightly degrades penetration and foot-sliding metrics while maintaining a similar CLIP Score, suggesting that these constraints mainly serve to refine geometric plausibility.

4.3 Qualitative Results

We further provide qualitative results to analyze the generated interaction quality and adaptive behavior. Fig. 3 presents qualitative comparisons between **ZeroCOI** and baseline methods under identical text prompts. The baseline methods tend to produce either overly conservative motions with limited interaction extent, or motions that exhibit mismatches between the intended semantics and the executed interactions. These artifacts often manifest as insufficient object engagement, noticeable penetration, or unstable foot contacts. In contrast, ZeroCOI generates more expressive interaction

Method	CLIP Score↑	CLIP Consistency↑	MD↑	MD_{local} ↑	$Pene_{\%obj}$ ↓	$Pene_{mean}$ ↓	$Pene_{max}$ ↓	FS↓
DAViD [Kim et al. 2025]	$24.3402^{\pm.029}$	$0.9650^{\pm.002}$	$\mathbf{0.6024}^{\pm.041}$	$0.1915^{\pm.023}$	$0.1222^{\pm.014}$	$0.1363^{\pm.016}$	$0.2458^{\pm.004}$	$0.5742^{\pm.005}$
TSTMotion [Guo et al. 2025]	$21.8726^{\pm.089}$	$0.9526^{\pm.000}$	$0.4612^{\pm.035}$	$0.0672^{\pm.005}$	$0.1507^{\pm.018}$	$0.2997^{\pm.008}$	$0.5125^{\pm.010}$	$0.6817^{\pm.006}$
AnimateAnyMesh [Wu et al. 2025]	$20.8474^{\pm.125}$	$\mathbf{0.9945}^{\pm.000}$	-	-	-	-	-	-
ZeroCOI (Ours)	$\mathbf{28.3522}^{\pm.058}$	$0.9848^{\pm.000}$	$0.4054^{\pm.021}$	$\mathbf{0.2794}^{\pm.018}$	$\mathbf{0.0764}^{\pm.011}$	$\mathbf{0.0832}^{\pm.001}$	$\mathbf{0.0957}^{\pm.003}$	$\mathbf{0.0416}^{\pm.008}$

Table 2. **Generalization to Unseen Objects.** Quantitative comparison on the held-out unseen-object set, demonstrating robust semantic alignment and physical plausibility under distribution shifts in object category and geometry. All results are averaged over five runs, with 95% confidence intervals shown.

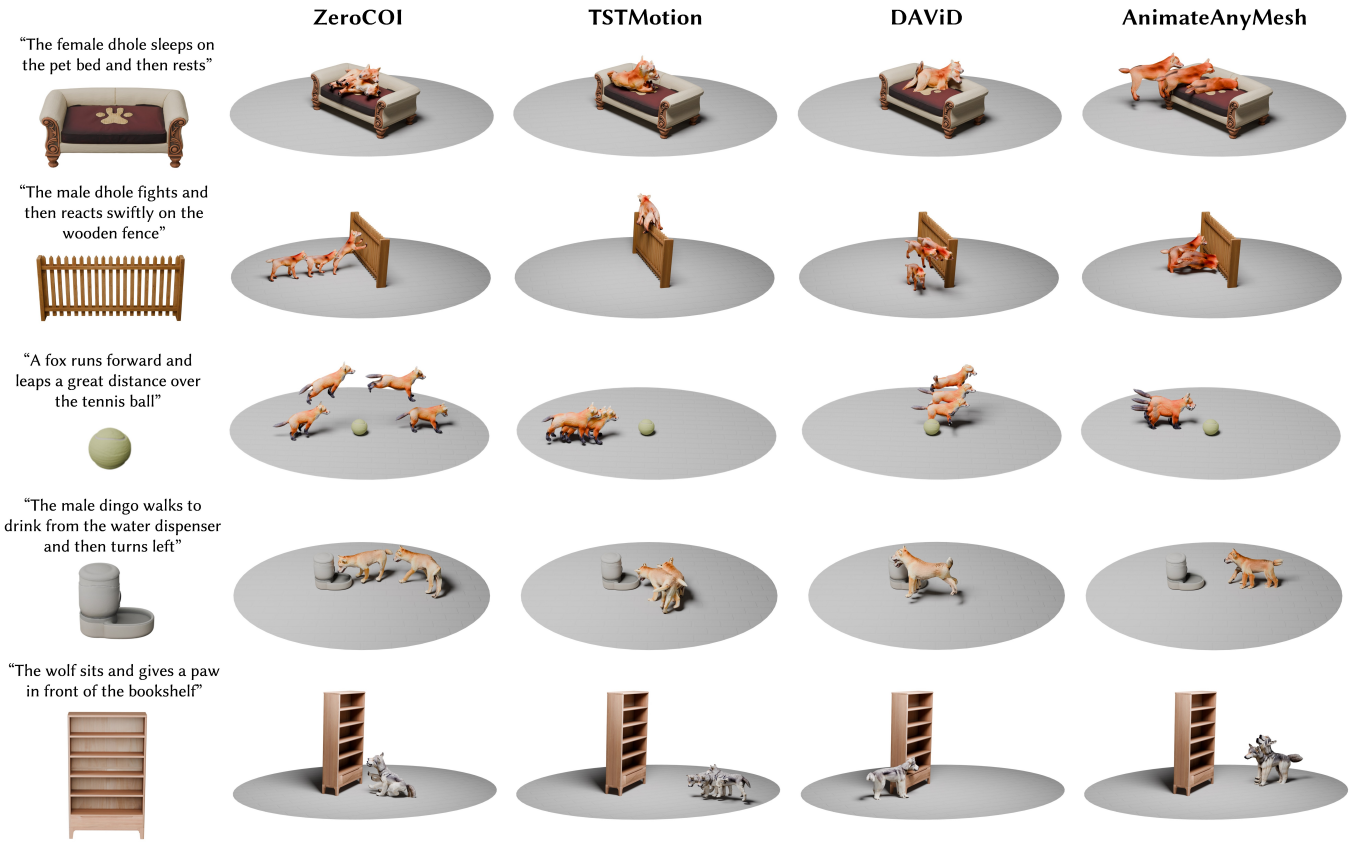


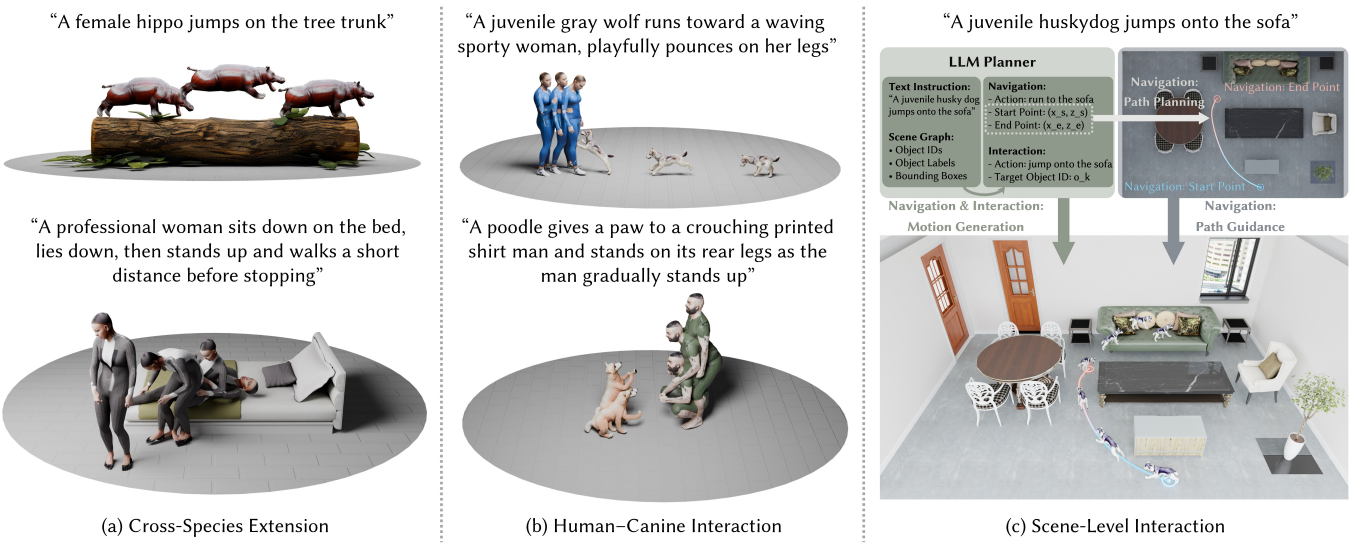
Fig. 5. **Generalization to unseen objects. ZeroCOI** synthesizes semantically consistent and physically plausible COIs with objects not observed during training, where the **top row** shows interactions with unseen object instances, and the **remaining rows** involve objects from unseen categories. The generated interactions remain consistent with the textual instructions and adapt to object geometries, demonstrating robust generalization beyond memorized objects.

behaviors that better align with the semantic intent of the input text, while maintaining stable contacts and physically plausible motion trajectories throughout the interaction.

Fig. 4 further illustrates **ZeroCOI** 's adaptive and diverse interaction behaviors under controlled settings. Specifically, we evaluate (i) the same canine interacting with objects of different heights and geometries, (ii) different canines with varying body shapes interacting with the same object, and (iii) multiple motion realizations under the same prompt and object condition. In the first two cases, **ZeroCOI** adapts interaction execution to the canine's morphology and the object's geometric properties, exhibiting continuous behavioral adaptation rather than reproducing fixed or memorized interaction patterns. In the third case, **ZeroCOI** exhibits meaningful interaction diversity, producing distinct motion realizations that vary in either internal motion dynamics or valid interaction trajectories, while remaining semantically consistent with the prompt.

4.4 Generalization

We evaluate generalization on a held-out unseen-object set, with results summarized in Tab. 2. **ZeroCOI** exhibits trends highly consistent with the main quantitative evaluation, suggesting a robust capability to synthesize semantically aligned and physically plausible canine–object interactions even under distribution shifts in



(a) Cross-Species Extension (b) Human–Canine Interaction (c) Scene-Level Interaction

Fig. 6. **Extended applications of our framework.** We present three extended application scenarios: (a) cross-species extension, including human and hippopotamus examples; (b) human–canine interaction with dynamic human motions; and (c) scene-level interaction with obstacle-aware navigation and target object interaction.

object category and geometry. Despite the increased geometric complexity and rarity of unseen objects (e.g., bookshelves, pet beds, water dispensers, and fences), **ZeroCOI** maintains the highest *CLIP Score* while consistently achieving the lowest penetration and *Foot Sliding* across all methods. These results indicate that **ZeroCOI** generalizes effectively in both semantic alignment and physical reasoning, rather than overfitting to object instances observed during training. Fig.5 further presents qualitative results generated with unseen objects. **ZeroCOI** produces coherent interaction behaviors that respect object affordances and geometric constraints, confirming that the learned interaction distribution extends beyond the training set and remains stable under previously unseen object geometries.

5 Extended Applications

We demonstrate **ZeroCOI**'s scalability via three application scenarios, including (i) Cross-Species Extension, (ii) Human–Canine Interaction, and (iii) Scene-Level Interaction.

Cross-Species Extension. Beyond canines, our framework can be adapted to other articulated species by replacing only the motion dataset and the underlying parametric body model. Specifically, we instantiate the pipeline with SMPL for humans and the standard SMAL model for non-canine quadrupeds. Since our framework already conditions motion generation on shape parameters β, it can naturally accommodate SMPL humans with different body shapes, as well as diverse SMAL-based quadrupeds with varying morphologies. Fig. 6(a) presents human and hippopotamus examples, showing that our framework can synthesize semantically correct and physically plausible interaction motions for species with different skeletal structures and body shapes.

Human–Canine Interaction. For this application, we treat the human motion as a time-varying mesh sequence, replacing the static object mesh in the original pipeline. The object BPS representation is correspondingly extended to a frame-wise form, where the human geometry feature and the canine-to-human interaction feature are computed at each motion frame. Following the same object-aware semantic augmentation strategy, we replace the object category label with the human motion description, and compose it with the motion-centered canine prompt to obtain a human-canine interaction text. Fig. 6(b) shows representative human–canine interaction results, where the generated canine motions respond to dynamic human actions and remain semantically consistent with the text prompts. These examples demonstrate the effectiveness of our framework in extending to dynamic interaction targets, while preserving coherent and physically plausible canine motions.

Scene-Level Interaction. We extend our framework to scene-level interaction by introducing a high-level LLM planner and obstacle-aware path guidance. Given a text instruction and a scene graph, the LLM planner grounds the instruction to a target object instance, decomposes the task into navigation and interaction subgoals (*i.e.*, approaching the target object and performing the intended interaction), and specifies the start and end points for navigation. A* path planning is then applied to obtain a 2D collision-free navigation path between them. We generate motions for the navigation and interaction stages separately. To achieve obstacle avoidance, we guide the navigation-stage sampling using the distance between the world-space root trajectory and the planned path. To ensure temporal continuity between the two stages, we further apply continuation guidance to the interaction-stage sampling, encouraging its initial motion window to match the terminal window of the navigation motion in terms of key joint positions, rotations, and velocities. Fig. 6(c) shows the resulting scene-level motion with obstacle-aware navigation and target object interaction.

6 Conclusion

We present **ZeroCOI**, a zero-shot learning framework that synthesizes canine–object interactions from text prompts by leveraging semantic and physical priors in large-scale pre-trained text-to-video diffusion models, without paired text–object–motion supervision. Extensive evaluations show that **ZeroCOI** improves both semantic alignment and physical plausibility, and generalizes robustly to unseen object categories and geometries. Furthermore, extended applications illustrate the scalability of **ZeroCOI** across various scenarios, including cross-species extension, human-canine interaction, and scene-level interaction. More broadly, **ZeroCOI** points toward a scalable paradigm for language-driven synthesis of canine interactions, potentially reducing reliance on manual animation authoring or extensive motion capture. We believe this direction can facilitate richer canine behaviors in interactive applications such as games, virtual reality, and digital content creation.

Full related work, implementation details, dataset statistics, limitations, and additional results are provided in the appendix.

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